

LionWeb: Status + Use cases

LangDev 2026
Meinte Boersma, Niko Stotz

Short introduction



Mission



To create an ecosystem of interoperable components for building language-oriented modeling tools on the web.

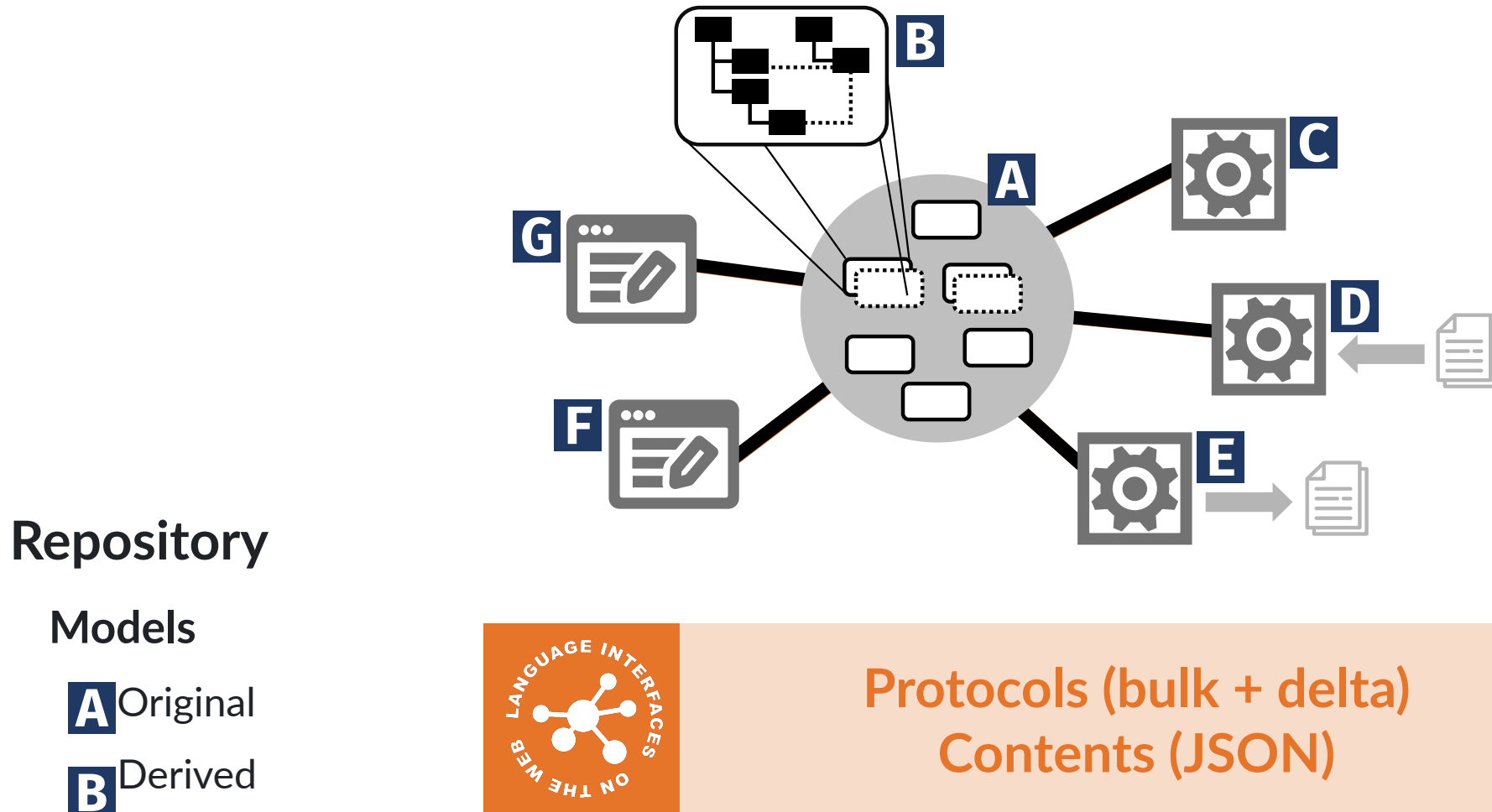
Adopt language engineering

Reduce vendor lock-in

Mix and match components

Foster innovation and incremental improvements

Reference Architecture



Clients

Editors

- F** Visual
- G** Projectional

Processors

- C** Model checker
- D** Importer
- E** Generator

Talks



- LangDev 2023 ([slides](#), [video](#))
- LangDev 2024 ([slides](#), [video](#))
- SEN Symposium 2024 ([conference](#), [slides](#))
- MODELS 2024 ([conference](#))
- LangDev 2025 ([slides](#))
- MBSE 2025 (slides, [conference](#))
- MODELS 2026 — Industry Day (slides <== TODO)

Other talks about LionWeb @ LangDev'26

Title	Speaker	Affiliation
<u>LionWeb in the Real World: Lessons from a Massive Legacy Migration</u> Thursday 17:15	Alessio Stalla	Strumenta
<u>Reinvigorating TRUMPF laser programming with LionWeb</u> Friday 8:30	Niko Stotz, Johannes Tauber	F1RE, TRUMPF
<u>Implementing MyFPL with LionWeb</u> Friday 11:30	Meinte Boersma	DSL Consultancy

Status



Progress

New core members

- Sergey Zverlov (F1RE)

Office hours (for outreach)

- Once a month (every second Friday) — successful!

Progress

- Released 2026.1 (as of now!)
 - Main feature: Delta Protocol ([specification](#))
- [OpenAPI specification for Bulk API](#)

Delta protocol

Released as part of 2026.1

Is seeing heavy use *in production* with a large customer²

- Spec and implementations turned out to be a very good fit

Changes since last year:

- Command and event names more consistent

- Chunked messages

- Precisely-defined errors

Implementations available in: [C#](#), [TypeScript](#), [JVM](#), and [server](#)

Use cases



Overview

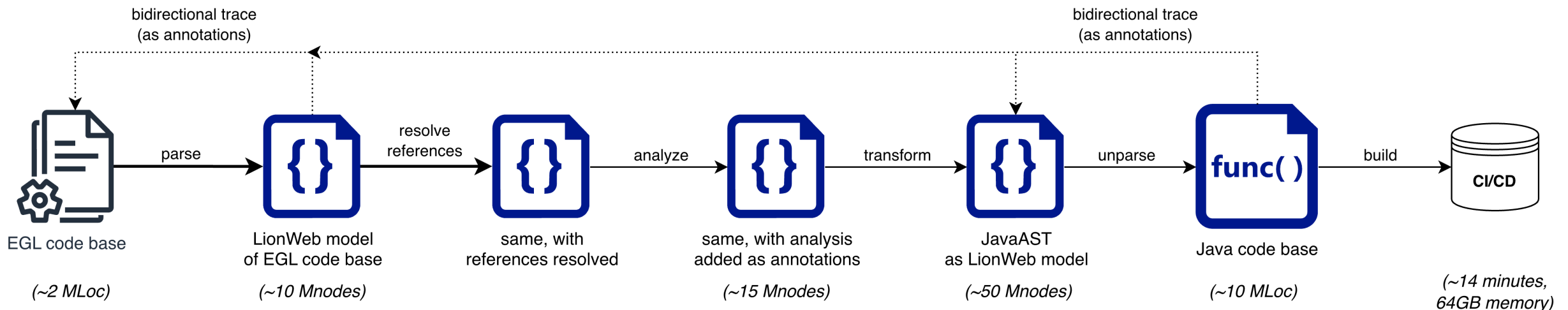
1. Strumenta: legacy transformation of huge code bases
2. TRUMPF: essential part of infrastructure
3. LionWeb implementation of an MCP
4. LionWeb as integration platform
5. How to implement an LWB with LionWeb



Separate talk, today (Thursday) @ 17:15:

[LionWeb in the Real World: Lessons from a Massive Legacy Migration](#)

Use case: legacy transformation of *huge* code bases
e.g. ~2 MLoC of IBM EGL code





LionWeb-specific optimizations:

- Save each partition as a separate file in a custom Protobuf variant of the LionWeb serialization format
- Intermediate results are stored as annotations: avoids recomputation
- Cache things like “what are *all* (incl. inherited) features of this classifier?”
- Plenty of “old-fashioned Java optimizations” brought to `lionweb-jvm`
- Parallelize: parsing files, processing partitions



Separate talk, tomorrow (Friday) @ 8:30:

[Reinvigorating TRUMPF laser programming with LionWeb](#)

Use case: a laser CAM system with its application data held as a LionWeb model that's replicated across multiple components

Key takeaways:

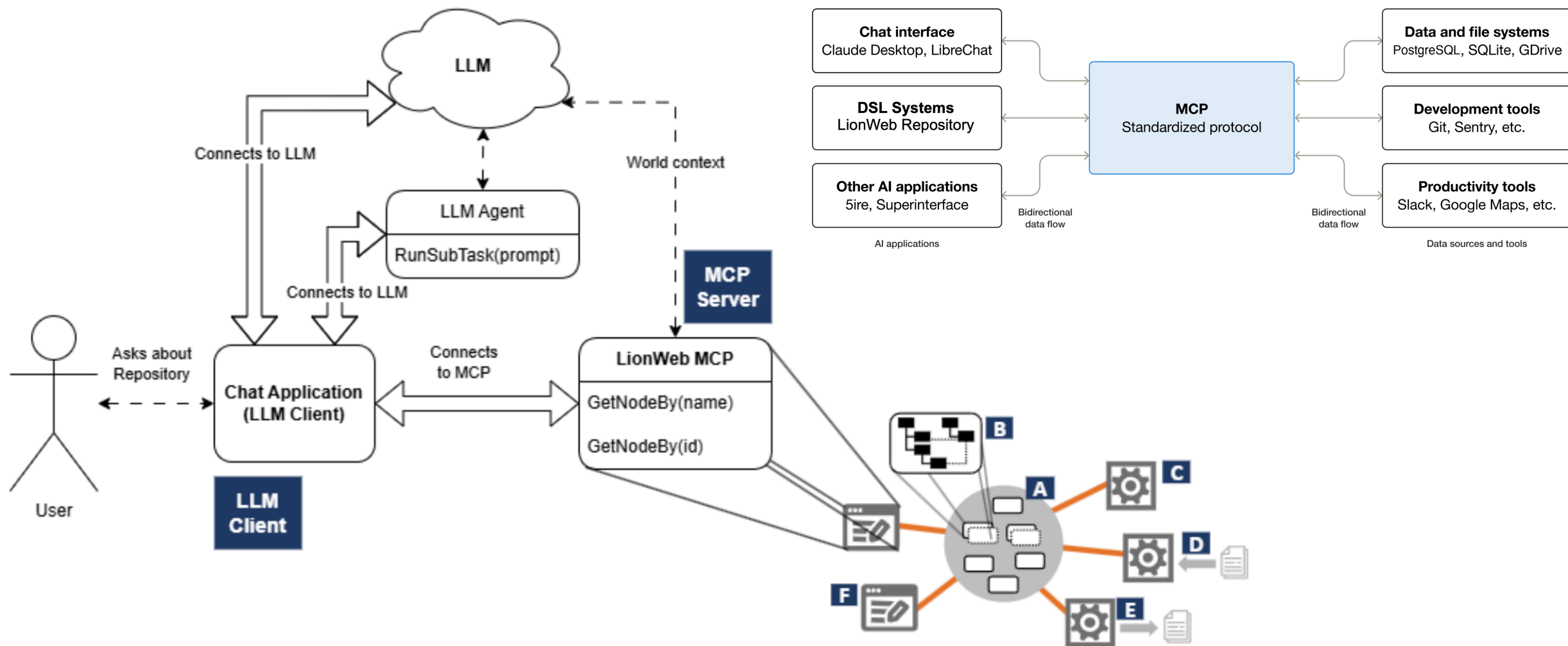
- The project turned out to need most of LionWeb's capabilities (Delta, Bulk, etc.)
- Delta protocol is used as a mechanism to sync a replicated model
- TRUMPF funded the LionWeb open source work
 - especially the C# and TypeScript implementations

LionWeb MCP

Atanas Marchev (F1RE) built a LionWeb MCP server:

- Exposes models in MPS as LionWeb repositories through MCP tools
- These MCP tools are LionWeb-specific, but aspecific to specific LionWeb languages.
Examples:
 - `findNodeByName`
 - `changePropertyValue`
- Uses a LLM-powered chatbot to interact with the repositories
- Demoed: interrogate and manipulate RegelSpraak model in ALEF

LionWeb MCP



LionWeb-driven integration platform

Niko?

LionWeb-driven LWB

Niko?

Team and Contact



Team

Meinte Boersma	DSL Consultancy	Alex Shatalin	JetBrains (MPS)
Erkan Diken	Belastingdienst	Niko Stotz	F1RE
Norman Koester	itemis (Modelix)	Federico Tomassetti	Strumenta
Sergej Koscejev	Freelancer	Markus Voelter	Freelancer
Sascha Lisson	itemis (Modelix)	Ulyana Tikhonova	F1RE
Pedro Molina	Metadev	Jos Warmer	Freelancer (Freon)
Eugen Schindler	Canon Production Printing	Sergey Zverlov	F1RE

Contact

Want to work with us:

info@lionweb.io

GitHub:

<https://github.com/LionWeb-io>

Web:

<https://lionweb.io/>

Slack to follow us, use, or implement LionWeb:

https://join.slack.com/t/lionweb/shared_invite/zt-2k1k5bsv7-EvC2IABlgSxWrqJNVOZ0HQ

